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Cybersecurity and Sustainable Development – Bibliometric Study

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Abstract

The aim of this article is to explore the interdependencies between cybersecurity and sustainable development, to map the scientific landscape at their intersection using bibliometric techniques, and to identify emerging key research gaps. To investigate how this relationship has been addressed in the scholarly literature, a bibliometric review was conducted on selected 425 publications that include the terms 'cybersecurity' and 'sustainable development'. The article presents a broad comparison of the Scopus and Web of Science databases, with a justification for choosing the Scopus database. Co-citation, keyword-co-occurrence and thematic-evolution analyses carried out with VOSviewer and Bibliometrix, respectively, allowed to identify four principal research themes: (1) smart-energy infrastructure and technical security, (2) cyber-resilient critical systems, (3) ESG-oriented digital management and (4) sustainable industrial transformation. The primary novelty of this study lies in its contributions aligned with the research questions: (RQ1) the exploration of the connections between the terms 'cybersecurity' and 'sustainable development'; and (RQ2) the widening and presentation of the concept of green cybersecurity, defined as the deliberate design, deployment, and security solutions management that uphold environmental protection without compromising system resilience. The future research can focus on the multidisciplinary research agenda and strategy recommendations are then articulated to align ICT management and green jobs creation with circular-economy principles. Therefore, this study identified two relatively unexplored domains: (I) life-cycle-aware security design and (II) ultra-low-carbon cybersecurity.

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Keywords: digital resilience; green cybersecurity; green jobs; sustainable competitive advantage; sustainable strategic management

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1. Introduction

Digitalization supports climate monitoring, smart-energy management and circular-economy logistics, yet simultaneously enlarges demand for energy, critical minerals and resource-intensive infrastructure. However, their implementation has been found to raise the demand for energy, critical minerals, and infrastructure that is resource-intensive and dependent on natural environmental assets. This development paradox has been understood to create a structural tension between cybersecurity, which is required to safeguard digital systems, and sustainable development, which aims to minimize ecological and social harm. Problems and threats resulting from the unsustainable extraction and consumption of natural resources have led to the development of the concept of sustainable development, which aims to ensure the fulfilment of basic human needs - such as access to safe food, clean water, and adequate living standards - while remaining within planetary boundaries [1]. Parallel to concerns about sustainability, the exponential growth of Information and Communication Technologies (ICT) has been accompanied by a corresponding increase in cybersecurity threats [2]. These threats now extend beyond digital systems, posing risks to human safety and ecological integrity. Both the advancement of computer science and the principles of sustainable development exert significant influence on contemporary business practices [3]. Consequently, cybersecurity has emerged as a core strategic concern often related to the sustainable strategies', increasingly embedded in corporate management and operational decision-making [4]. Security, therefore, should be conceptualized in dual terms: (a) as the protection of digital infrastructure and (b) as the safeguarding of the environmental and social systems upon which that infrastructure ultimately depends.

The aim of this article is to explore the interdependencies between cybersecurity and sustainable development, to map the scientific landscape at their intersection using bibliometric techniques, and to identify emerging key research gaps. To achieve this, the Scopus scientific database was queried using keywords designed to balance precision and research reproducibility [5]. An initial query - comprising the terms *cybersecurity* and *sustainable development* - was employed to explore the Scopus database. Following an analysis of relevant synonyms (in literature review section), it was decided to exclude 'sustainability' as a keyword in the research query, given its conceptual overlap with 'sustainable development', within the scope of the Scopus database search. Sustainability is a broader and more abstract idea, often used in diverse disciplinary contexts, whereas sustainable development refers to a more structured and goal-oriented framework guided by policy instruments [1]. This exclusion was methodologically justified to ensure terminological precision and to avoid the dilution of results due to the broader and more ambiguous usage of 'sustainability' across disciplines. Based on the retrieved dataset, co-citation, keyword co-occurrence, and thematic evolution analyses were conducted using VOSviewer and the Bibliometrix R-package, respectively. While this methodological approach enables the identification of current research trends and major thematic clusters, it is inherently constrained by the specificity of the selected keywords [6]. To mitigate this limitation, widely recognized and frequently occurring terms were intentionally used to enhance comparability across studies. Accordingly, the study addresses the following research questions (RQs):

RQ1: What are the dominant research themes linking cybersecurity and sustainable development?

RQ2: How can the concept of green cybersecurity be expanded and defined in academic literature to integrate both cybersecurity and sustainable development?

As a result, by employing bibliometric mapping, this article investigates the frequency and contextual co-occurrence of key concepts, including cybersecurity, sustainability, digital resilience, green ICT, and cyber-physical systems [7]. The analysis uncovers several inconsistencies and underexplored areas that present promising avenues for future research [9]. One particularly salient issue is the paradox of green or sustainable cybersecurity - that is, how to ensure robust digital protection without generating an excessive environmental footprint. Another tension arises between the demand for high availability and uninterrupted functioning of digital systems and the principles of circular-economy design, such as modularity and component interchangeability [8]. At the organisational level, a weak coupling between ICT management frameworks and sustainable strategies continues to hinder the integration of digital-footprint metrics into environmental and social performance assessments.

Unlike previous reviews that treat cybersecurity and sustainability as largely separate domains, this paper is the bibliometric analysis to frame "green cybersecurity" as an emergent, interdisciplinary research field. The article clearly demonstrates bibliometric triangulation through the use of two distinct analytical tools: VOSviewer and Bibliometrix, which in the literature is recognised as a form of methodological triangulation within bibliometric

analysis [10]. By combining co-citation, thematic-evolution, and overlay-visualisation techniques, the study not only charts the intellectual structure of the field but also pinpoints two neglected domains (I) life-cycle-aware security design and (II) ultra-low-carbon cybersecurity; thereby providing a clear research agenda for scholars and practitioners.

The article is organised as follows: following this introduction, which provides the rationale for addressing the topic, Section 2 presents a literature review, beginning with an overview of the interdependencies between cybersecurity and sustainable development, followed by the presentation of key concepts and their commonly used synonyms. Section 3 outlines the research methodology, including the bibliometric mapping based on a literature review supported by the VOSviewer software and Scopus database. The analysis was further extended using the Bibliometrix R-package, with results presented in the form of co-citation, keyword co-occurrence, and thematic-evolution analyses - summarised graphically. Section 4 discusses the identified research gaps, as well as the policy implications and managerial relevance of the findings. Finally, Section 5 concludes the article by outlining its limitations and suggesting directions for future research.

2. Literature

The interdependencies between cybersecurity and sustainable development have been acknowledged with growing frequency by both scholars and practitioners [8]. Whereas security was formerly located almost exclusively within the technical domain of computer science, recent scholarship has shown that sustainability principles [11] particularly those relating to energy, water and critical-material efficiency - ought to guide every stage of security design, implementation and management. These interdependencies extend beyond the ICT sector, permeating energy and critical infrastructure, advanced manufacturing, transport and logistics, public administration, commerce, finance, healthcare, education, telecommunications and agribusiness [12,13]. Across these sectors a structural tension has been observed. Round-the-clock protection of digital assets requires continuous server uptime, intensive cooling and regular hardware replacement, activities that consume energy and water and generate electronic waste [14]. Simultaneously, the transition to low-carbon business models demands strict resource protection and circular-economy principles, such as modularity and component interchangeability [15]. Failure to reconcile these competing pressures results in substantial environmental costs, including heightened pollution and greenhouse-gas emissions.

A growing body of literature argues that the infusion of green values into security practice can provide competitive advantage [12]. Enterprises embedding green cybersecurity within their management frameworks have been found to signal environmental credibility to investors, customers and regulators; lower operating costs through energy-efficient architectures; and unlock new revenue streams in sustainability-oriented markets. This alignment also fosters green innovation and the creation of green jobs [16] - positions that blend security expertise with life-cycle assessment, renewable-energy integration or low-carbon software engineering [17]. Such roles contribute directly to the reduction of anthropogenic pressure on ecosystems and to the attainment of digital resilience, defined as the capacity to maintain secure, reliable digital operations while minimising negative externalities.

To ground the subsequent analysis, definitions and commonly used synonyms for the two focal constructs 2.1 cybersecurity and 2.2 sustainable development were reviewed. Their frequency and contextual co-occurrence in academic discourse are examined with VOSviewer, drawing upon Scopus-derived data. These steps illustrate the conceptual evolution of the field and prepares the way for evaluating how sustainable strategic-management frameworks may leverage green cybersecurity to enhance long-term competitiveness and societal value.

2.1 Cybersecurity – definition and synonyms

Cybersecurity is most commonly understood as the ensemble of policies, procedures, and technologies used to shield computer systems, networks, and data from unauthorised access, disruption, or damage [18]. Cybersecurity can be widened to cover the protection of both digital and physical assets across three interdependent and connected domains: 1) the natural environment; 2) the continuity of economic processes; 3) the health and property of society at large. Cybersecurity functions as a multi-level integrative mechanism that links ICT management with environmental-management systems and other strategic functions in organisations [19].

The green cybersecurity concept comprises two complementary elements: (1) an integrative dimension that embeds secure, low-carbon digital infrastructures within broader "green" value chains, creating safe and sustainable interdependencies; and (2) a design dimension that applies life-cycle thinking to hardware and software, favouring modularity, repairability, and eventual reuse rather than discarding them. By coupling these elements, the protective layer required for virtual operations can be delivered without intensifying anthropogenic pressures on natural systems. Cybersecurity meaning evolved from data protection to wider, social contexts. This evolution towards different meanings of cybersecurity terms is presented in Table 1. A bibliometric review performed in VOSviewer and Scopus database shows how a variety of synonymous and related terms have gained prominence, illustrating the field's dynamism and its increasing entanglement with sustainability discourse.

Table 1. Synonyms for "Cybersecurity".

Term (First recorded use - year)	Frequency of use	Remarks
Information security (1967)	High	A broad term encompassing all aspects of information protection.
IT security (1990)	High	Focuses on the security of information technologies.
Computer security (1970)	Medium	Emphasizes the protection of computer systems.
Network security (1957)	Medium	Pertains to safeguarding computer networks.
Data security (1961)	Medium	Concentrates on protecting data from unauthorized access.
Cyber defence (1992)	Low	Commonly used in military and governmental contexts.

Source: Own elaboration

An analysis conducted in the Scopus database examined when scholarly interest in these terms first emerged, identifying the earliest publications on the topic. In addition to tracing the origins of interest, the study also reviewed the most frequently cited articles associated with selected synonyms for cybersecurity. The most influential and highly cited works tend to be interdisciplinary in nature, transferring conceptual constructs across academic domains. This highlights not only the historical depth of certain terms but also their adaptability to evolving scientific problems and sector-specific discourse. The keywords identified in Table 1 are long-established, yet they continue to characterize specific fields or sectors [20,21]. Consequently, the frequency of their usage varies and is directly linked to current research challenges.

2.2 Sustainable development – definition and synonyms

Sustainable development is a concept of progress that meets the needs of the present generation without compromising the ability of future generations to meet their own needs. This definition was first articulated in the 1987 Brundtland Report, formally titled "Our Common Future" [1].

Table 2. Synonyms for "Sustainable Development"

Term (First recorded use - year)	Frequency of use	Remarks
Sustainable development (1929)	High	The most commonly used term in scientific literature
Environmental sustainability (1905)	High	Focuses specifically on the long-term viability of natural ecosystems.
Sustainable growth (1959)	Medium	Often applied in economic and business contexts.
Green development (1976)	Medium	Emphasizes ecological and environmental considerations.
Lasting development (1990)	Low	A less common term in academic discourse.

Source: Own elaboration

Keywords that appear for the first time in the academic literature do not necessarily correspond with high

citation counts, even when a given publication is credited with introducing the term. This observation suggests that early conceptual formulation, in isolation, does not ensure scholarly impact. Instead, the interdisciplinary transfer of a concept - such as the migration of a term from environmental science into policy discourse - and its subsequent adoption as a political or strategic instrument may be essential for its broader recognition and citation. Conversely, synonyms that are used infrequently often lack grounding within a specific disciplinary context, which may hinder their uptake. Such terms are frequently perceived as ambiguous or imprecise.

3. Methods

This study employs a bibliometric approach to identify research trends, conceptual overlaps, and emerging gaps at the intersection of cybersecurity and sustainable development [22]. The Scopus database was used as the primary source for data collection due to its comprehensive coverage of peer-reviewed scientific literature. Table 3 presents a rationale for choosing Scopus over Web of Science. Scopus offers broader coverage, both in terms of the number of indexed publications and the diversity of disciplines represented. The decision to use Scopus was further supported by its high degree of interdisciplinarity, greater availability of open access content, and stronger representation of emerging and flexible scientific fields. These characteristics make Scopus particularly well-suited for studies situated at the intersection of traditionally distinct domains, such as cybersecurity and sustainable development. This is important because the nexus between these terms remains underexplored.

Table 3. Comparison of Scopus and Web of Science Databases

Feature	Scopus	Web of Science
Publisher	Elsevier	Clarivate Analytics
Number of Journals	Approx. 22,800	Approx. 13,100 (20,556 including ESCI)
Subject Coverage	Natural sciences, medical sciences, social sciences, and humanities	Natural sciences, social sciences, arts, and humanities
Coverage Period	Ongoing indexing since 1970 (with some retrospective access)	Since 1945 (expandable to 1900)
Geographic Reach	Global, with strong representation of non-U.S. publications	Primarily focused on U.S. and European publications
Interdisciplinarity	High	Moderate
Open Access Availability	Greater number of open access resources	Lower open access availability
Bibliometric Tools	Advanced (CiteScore, h-index, Scopus AI)	Advanced (Impact Factor, Journal Citation Reports, Web of Science Research Assistant)

Source: Own elaboration

Both databases are widely used in bibliometric analyses and offer distinct tools for evaluating the impact of scientific publications. Scopus stands out for its user-friendly interface and integration with advanced bibliometric tools. It includes a broader range of titles from both local and international markets, many of which are omitted in Web of Science - particularly practical and interdisciplinary journals. Scopus also tends to be more current and adaptable in incorporating emerging fields such as project management, entrepreneurship, and digital management, whereas Web of Science is more conservative and slower in indexing new titles. Furthermore, Scopus provides better representation of research output from Europe, Asia, and Latin America, which is particularly important in international and cross-cultural management research. In contrast, Web of Science has a more “Anglo-Saxon” orientation, dominated by publications from the United States and the United Kingdom. Scopus also offers robust citation analysis tools, enabling users to track publication trends, co-authorship patterns, and institutional affiliations - valuable features for evaluating scholarly output and shaping publication strategies. Notably, Scopus facilitates the identification of influential authors within niche areas of management studies, a task often more difficult in Web of Science.

In this research bibliometric mapping techniques were applied to visualise the co-occurrence and frequency of selected terms. Particular attention was paid to the historical emergence of these terms, their first recorded appearances, and the citation performance of publications that introduced or popularised them. A notable challenge in this process stemmed from the semantic ambiguity and interdisciplinary transfer of certain keywords. Terms often carry different meanings across disciplinary contexts, complicating interpretation. This underscores the importance of careful keyword selection, where clarity and disciplinary specificity are essential. This procedure allowed the generation of the Bibliometrix chart presented in Fig. 1.

The bibliometric analysis conducted in this study followed a four-step process:

1. Query formulation - aimed at capturing interdisciplinary intersections.
2. Export of bibliometric data - based on extraction of key indicators such as the number of publications, citation counts, and term co-occurrence.
3. Data mapping using analytical tools - visualization and cluster analysis using software such as VOSviewer and Bibliometrix, enabling thematic grouping and structural mapping of the research field.
4. Identification of research gaps and underrepresented topics - comparative analysis of keyword frequency across different time periods to detect emerging trends and areas of low scholarly coverage.

The .csv file exported from Scopus (425 records, no time frame was specified) was imported into VOSviewer for keyword co-occurrence analysis with the full-counting option applied to all keywords. Default thresholds were retained for minimum keyword frequency and cluster resolution. This setting maximised coverage while avoiding manual bias in term selection.

A query, presented in Table 4, was constructed using the keywords cybersecurity and sustainable development, identified through a preliminary analysis of disciplinary terminology. The process of querying the Scopus scientific database using selected keywords highlighted the critical importance of database selection in bibliometric research.

Table 4. Query used in Scopus exploration

Query syntax	No. of Results
(TITLE-ABS-KEY ("cybersecurity") AND TITLE-ABS-KEY ("sustainable development")) AND (EXCLUDE (DOCTYPE , "tb"))	425

Source: Own elaboration

In the VOSviewer software, the appropriate type of analysis and counting method were selected. A co-occurrence analysis was performed using the full counting method. The analysis included all keywords, rather than limiting the scope to author keywords or indexed keywords. All other parameters were left unchanged, with the software's default settings applied. Accordingly, the bibliometric map was excluded from this publication, but a detailed account of the methodological procedure and resulting findings has been retained. Then the same initial data were used in the Bibliometrix to generate Callon's strategic diagram Fig. 1. Bibliometrix was employed as the visualization results from VOSviewer proved unsatisfactory without additional filtering or reduction of the keyword set. The methodology thus integrates quantitative analysis (term frequency, citation counts) with qualitative reflection on language use, aiming to provide a nuanced understanding of how terminology shapes and reflects research trajectories within both cybersecurity and sustainability domains.

4. Results

The results reflect the successive steps of the research procedure, which involved the use of VOSviewer and Bibliometrix, respectively. Due to the illegibility of the map generated in the VOSviewer software, it has not been included in this article. Therefore, Callon's strategic diagram was generated by highlighting "sustainable development" on the explored literature [24]. Fig. 1 shows the outcome of the thematic cluster analysis conducted using the Bibliometrix software. This allowed to bridge interdisciplinary insights into cybersecurity and sustainable development [23].

The resulting map presented on Fig. 1 reflects the interdisciplinary nature between cybersecurity and sustainable development in Bibliometrix software. Fig. 1 serves as a valuable foundation for identifying knowledge gaps, emerging trends, and potential areas for future inquiry within the scientific literature.

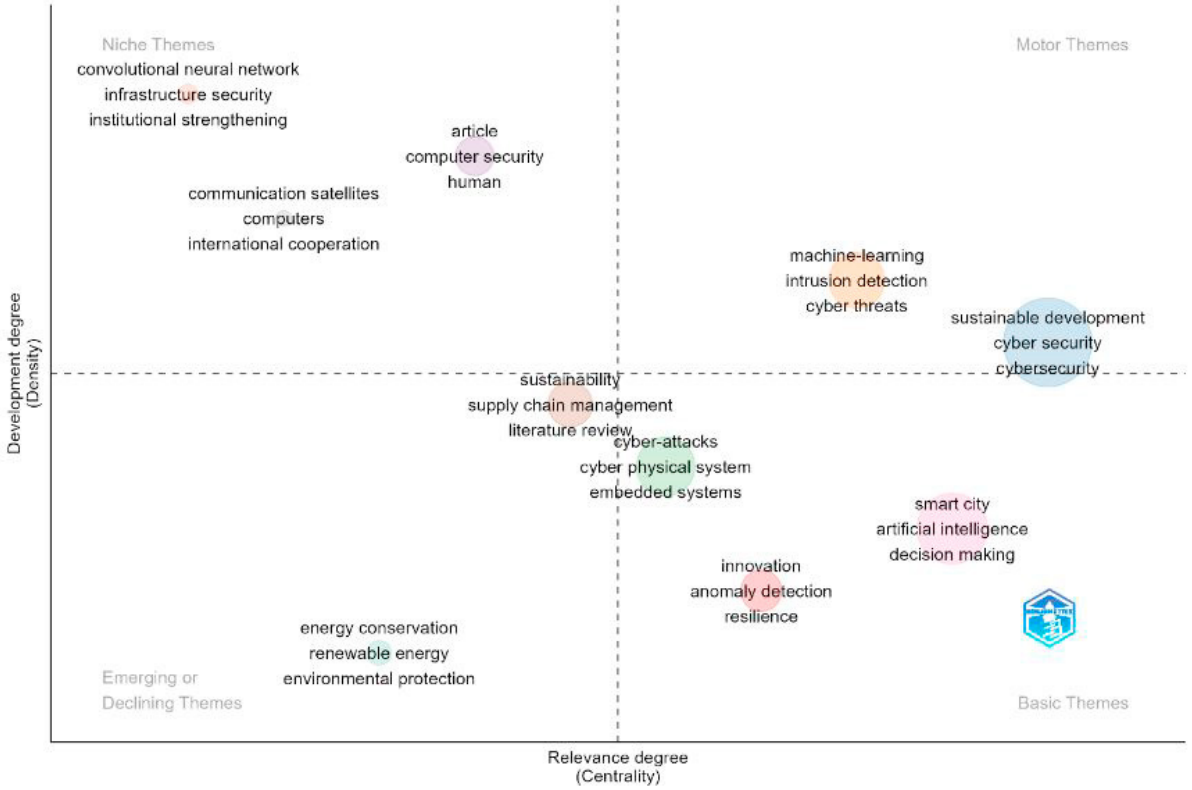


Fig. 1. Projects the clusters onto Callon's strategic diagram (x = centrality, y = density). Source: Own elaboration

On the Fig. 1 there are four research areas identified as: 1) motor themes (upper right) where sustainable development, cybersecurity, and a triad of machine learning, intrusion detection, cyber threats are presented, all of them are influential and conceptually mature; 2) niche themes (upper left) where terms of convolutional neural network, infrastructure security are cohesive but peripheral; 3) basic themes (lower right) such as: smart city, artificial intelligence, decision making, are central yet still consolidating; and 4) emerging or declining themes (lower left) are represented by energy conservation, renewable energy, environmental protection which show low maturity and peripheral relevance. Bubble size in Fig. 1 is proportional to keyword frequency in the initial set of extracted publications form Scopus, and colours differentiate the automatically generated clusters. Bibliometrix procedure allowed to identify four principal research themes: (I) life-cycle-aware security design, in which equipment reuse and modularity are embedded into hardware and software, and (II) ultra-low-carbon cybersecurity, comprising energy-efficient algorithms, cooling-optimised data centres and renewable-powered security services.

5. Discussion and Conclusion

This article presents an updated systematic bibliometric literature review of the cybersecurity - sustainability nexus based on Scopus data, extending previous studies by identifying underexplored subdomains and formalising the concept of green cybersecurity [25]. The analysis was built around the themes of sustainable development and cybersecurity. Conceptually, this study widens the definition of green cybersecurity. Practically, the findings offer insights for managers seeking to integrate cybersecurity with climate and sustainability agendas and provide guidance for researchers interested in green cybersecurity. The green cybersecurity serves two key functions:

protecting the mechanisms for resource redistribution and ensuring transparency to prevent greenwashing and the displacement of environmental externalities. Green cybersecurity can be defined as the intentional design and implementation of security measures that protect physical, digital, and financial assets essential to the ecological transition, while minimising their own environmental footprint. This perspective expands the conventional understanding of cybersecurity - typically confined to ensuring business continuity - by reframing it as a catalyst for long-term socio-ecological resilience.

The bibliometric analysis conducted in this study indicates that scholars are beginning to adopt this expanded conceptualisation of green cybersecurity as the emerging theme, and outcome of the interdisciplinary connections between cybersecurity and sustainable development presented in Fig. 1. Green cybersecurity offers a conceptual framework through which these regulatory instruments can be harmonised, enabling organisations to align digital protection with sustainability objectives [4]. The study builds upon existing bibliometric research while offering a more recent and targeted perspective on environmental aspects of cybersecurity, which were either marginally addressed or absent in earlier analyses. Nonetheless, several research gaps remain evident. These include the development of life-cycle-aware security design, the advancement of ultra-low-carbon cryptographic algorithms, the creation of ESG-aligned ICT performance metrics, and the implementation of cyber-secure circularity even within space systems. Addressing these gaps requires a shift from reactive, end-of-pipe risk mitigation strategies to proactive, sustainability-driven security design. Therefore, there were identified four principal research themes: (1) smart-energy infrastructure and technical security, (2) cyber-resilient critical systems, (3) ESG-oriented digital management and (4) sustainable industrial transformation.

The objective of the study was to map and critically synthesise the scholarly landscape at the intersection of cybersecurity and sustainable development and it has been fully achieved: a systematic bibliometric analysis of 425 Scopus-indexed publications was undertaken, four dominant research themes were delineated, two under-explored domains were identified, and the concept of green cybersecurity was formalised, thereby establishing a coherent foundation for future theoretical and policy-oriented work. Moreover, Research Questions 1 and 2 were explicitly addressed through the delineation of thematic clusters and the identification of research gaps.

While this study is based exclusively on Scopus data and English-language keyword searches, methodological triangulation was applied through the combined use of two bibliometric tools VOSviewer and Bibliometrix respectively, which employ distinct analytical logics and visualisation techniques. VOSviewer provided insight into keyword co-occurrence networks, while Bibliometrix enabled the analysis of co-citation structures and thematic evolution. This dual-tool approach strengthens the robustness and credibility of the findings, allowing for the identification of both central themes and emerging research frontiers. Nevertheless, bibliometric methods remain sensitive to keyword selection, language bias, and database indexing practices.

Furthermore, the exclusion of the term 'sustainability' from the query may limit the inclusion of certain interdisciplinary studies, and this limitation should be addressed in future replications by applying alternative or expanded keyword strategies. Future research should aim to validate and extend these findings by incorporating additional sources, including IEEE Xplore, non-English databases, and full-text natural language processing tools. Such approaches would allow for the inclusion of grey literature and region-specific scholarship that may otherwise be overlooked [26].

Building on the presented bibliometric analysis, future investigations should examine how green IT practices can be embedded within cybersecurity architectures so that the emerging field of green cybersecurity enhances digital resilience while simultaneously advancing sustainable development objectives articulated in ESG frameworks and circular economy models. Despite the growing consolidation of green cybersecurity as a research domain, significant gaps remain. Addressing these - particularly through the integration of circularity and carbon-awareness into the design of security systems - will be crucial to achieving genuinely sustainable digital infrastructures and advancing the broader objectives of a green digital economy.

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